



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: EE104Dept.: Electrical and Electronic EngineeringExam Duration: 120 minExam Paper Setter: Z. Gao & J. Ma

Question No.	1	2	3	4	5	6	7	8
Score								

This exam paper contains 8 questions and the score is 100 marks in total. (Please hand in your exam paper, answer sheet, and all your scrap paper to the proctor when the exam ends.)

Q1 (10 marks) In the circuit shown in Fig. 1, find the equivalent resistance between terminals a and b .

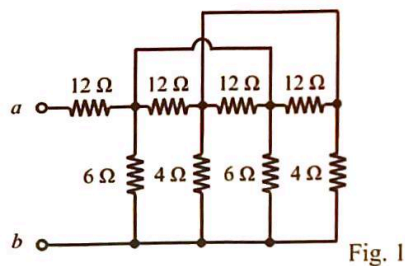


Fig. 1

Q2 (15 marks) Suppose you have only resistors with the following values in large quantities:

2.2Ω 20Ω 50Ω 62Ω

With different connection configuration of those resistors, how could you obtain the following resistances? Try to use the fewest number of resistors as you can. (Hint: Δ/Y circuits)

- a) 175Ω b) 32.1Ω c) 83.2Ω d) 119.2Ω e) 37.2Ω

Q3 (10 marks) In the circuit shown in Fig. 2, find v_s , R_1 , i_1

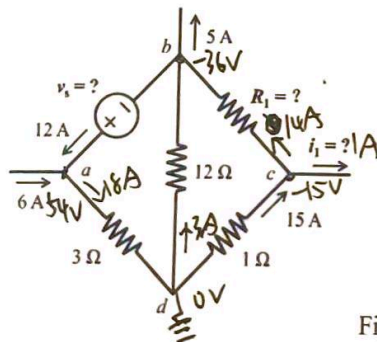


Fig. 2

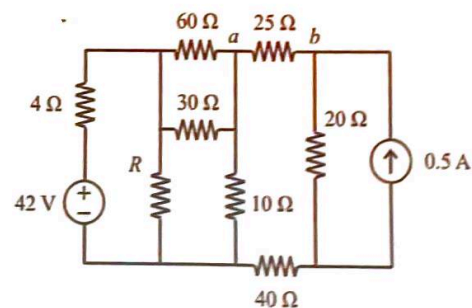


Fig. 3

Q4 (10 marks) In the circuit shown in Fig. 3, find the value of R if $v_{ab} = 0$ V.

Q5 (10 marks) Find the current i in the circuit shown in Fig. 4.

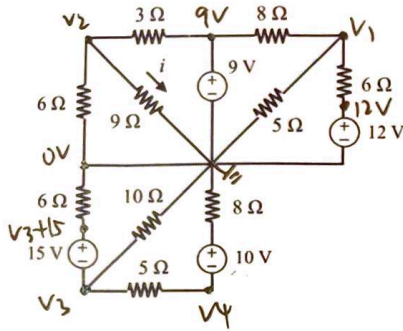


Fig. 4

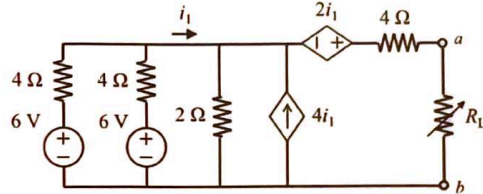


Fig. 5

Q6 (15 marks) In the circuit shown in Fig. 5, find (a) the Thévenin equivalent circuit between terminals a and b (for the circuit except for the R_L) and (b) the value of R_L at the maximum power transfer condition, and calculate the value of the maximum power.

Q7 (10 marks) Refer to the bridge amplifier shown in Fig. 6, determine the voltage gain v_o/v_i .

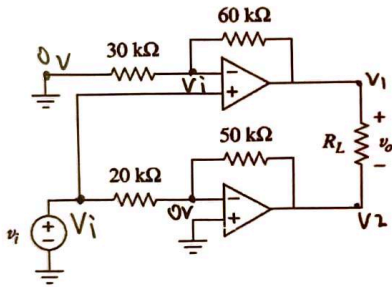


Fig. 6

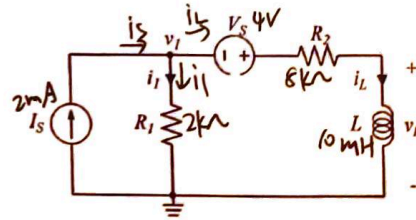


Fig. 7

Q8 (20 marks) Consider the circuit shown in Fig. 7 and assume that the inductor L has zero stored energy initially, i.e., $i_L(0) = 0$ A. Values of the other elements are $v_s = 4u(t)$ V, $i_s = 2u(t)$ mA, $R_1 = 2$ k Ω , $R_2 = 8$ k Ω , $L = 10$ mH. (Hint: Be careful about the units!)

- Derive the differential equation that describes the behavior of $i_L(t)$.
- Determine the time constant τ and the current $i_L(t)$.
- Determine the voltage $v_L(t)$.
- Determine the current $i_1(t)$.
- Determine voltage $v_1(t)$.